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EEE 202-1

LAB-01: First-Order RC Circuit's Response to Rectangular and Sinusoidal Pulses

Introduction:

The goal of this assignment is to find the transient response of a first-order RC circuit to both rectangular and sinusoidal pulses. The voltage across the capacitor will be analyzed for a periodic input of pulses. The values of the components that are required for the circuit will be gathered from a predetermined document on an individual level. The length and period of the pulses will be selected arbitrarily in a predefined range. For the preliminary section, analytical and simulative results will be gathered using MATLAB and LTSpice respectively and recorded. For the experimental section, the simulated circuit will be formed in real life and the same measurements that was taken in the preliminary section will be analyzed. The findings from both sections will be compared to see if the real-life measurements match our simulations.

Analysis:

The general analytical solution of the circuit (Figure A) will be derived using linear algebraic methods and by utilizing element parameters. The parameters of the resistor ($V = IR$) and the capacitor ($i_C = C\frac{dV_C}{dt}$) will be used.

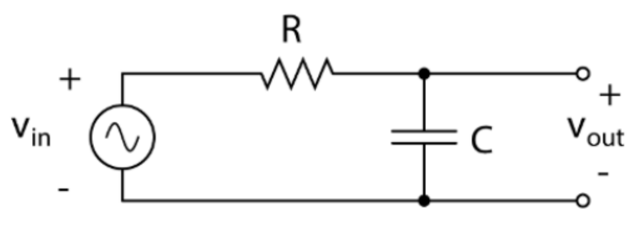

Figure A: Circuit that will be analyzed

Due to the fact that there is one loop, the current in said loop must be the same everywhere. The currents through the resistor and the capacitor can be expressed as:

$$i_R = \frac{V_{in}(t) - V_{out}(t)}{R} \quad (1) \qquad\qquad i_C = C\frac{dV_{out}(t)}{dt} \quad (2)$$

Since both of these currents are equal, we can equate both of them and rearrange the equation as:

$$\frac{dV_{out}(t)}{dt} + \frac{V_{out}(t)}{RC} = \frac{V_{in}(t)}{RC} \quad (3)$$

This is a first-order differential equation. By substituting the parameters in (3) into the general solution for non-homogenous first-order linear differential equations[1], we find the capacitor voltage to be:

$$V_{out}(t) = e^{-\frac{t}{RC}}\left(\frac{1}{RC}\int e^{\frac{t}{RC}}V_{in}(t)dt + C\right)V \quad (4)$$

where C is a constant that can be determined based on an initial condition. We will use this equation in the following parts to solve for $V_{out}(t)$ analytically.

Preliminary Part 1 (Rectangular Pulse Response):

First, the "Lab 1 RC assignment" document was checked to determine the value for the resistor and capacitor. These values were found to be 7kΩ and 560nF respectively. Through these values, the time constant was found to be 3.92 milliseconds. Since we have to pick T_0 (pulse period) and T_1 (pulse duration) based on a range of (30τ, 50τ) and (0.5τ, 3τ) respectively, 125 ms and 5 ms were chosen so that the signal frequencies would be integers in a decimal system while being in the given range. The rectangular pulse function, whose magnitude was given as 6 Volts, would act as a constant voltage with a magnitude of six for 0<t<T_1 and as zero volts for T_1<t<T_0. At the beginning of each repetition period, the voltage was assumed to be zero, therefore, we had the initial condition of $V_{out}(0) = 0V$.

Using these values and equation (4), the integral for the pulse duration was rewritten with time in milliseconds as:

$$V_{out}(t) = e^{-\frac{t}{3.92}}\left(\frac{1}{3.92}\int 6e^{\frac{t}{3.92}}\,dt + C\right)V = 6 - 6e^{-\frac{t}{3.92ms}}V \quad (5)$$

for the pulse duration, which is the time interval of 0<t<T_1.

For the interval of T_1<t<T_0, we can find the voltage at T_1, set it as the initial condition and re-evaluate the integral. Plugging in 5 ms into (5), we get approximately 4.3243 Volts. Since, during this interval, V_{in} is equal to zero volts, we can rewrite (4) as:

$$V_{out}(t) = e^{-\frac{t}{3.92}}\left(\frac{1}{3.92}\int 0e^{\frac{t}{3.92}}\,dt + C\right)V = 4.3243e^{-\frac{t-T_1}{3.92ms}}V \quad (6)$$

for the time between the end of the current pulse and the start of the next pulse, which is the time interval of T_1<t<T_0 where t is in milliseconds. Using these equations, we can model and plot the general function $V_{out}(t)$ by repeating the plot for 0<t<T_0 indefinitely, due to the assumption that the capacitor sufficiently discharges to zero volts before the start of a new rectangular pulse.

Using MATLAB, the function was plotted analytically (Figure B, Appendix). LTSpice was used to draw and simulate a transient circuit that fit our parameters (Figure C).

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[1] M. D. Greenberg, *Advanced Engineering Mathematics*, 2nd ed., Englewood Cliffs, NJ, USA: Prentice-Hall, p. 1, 1988.

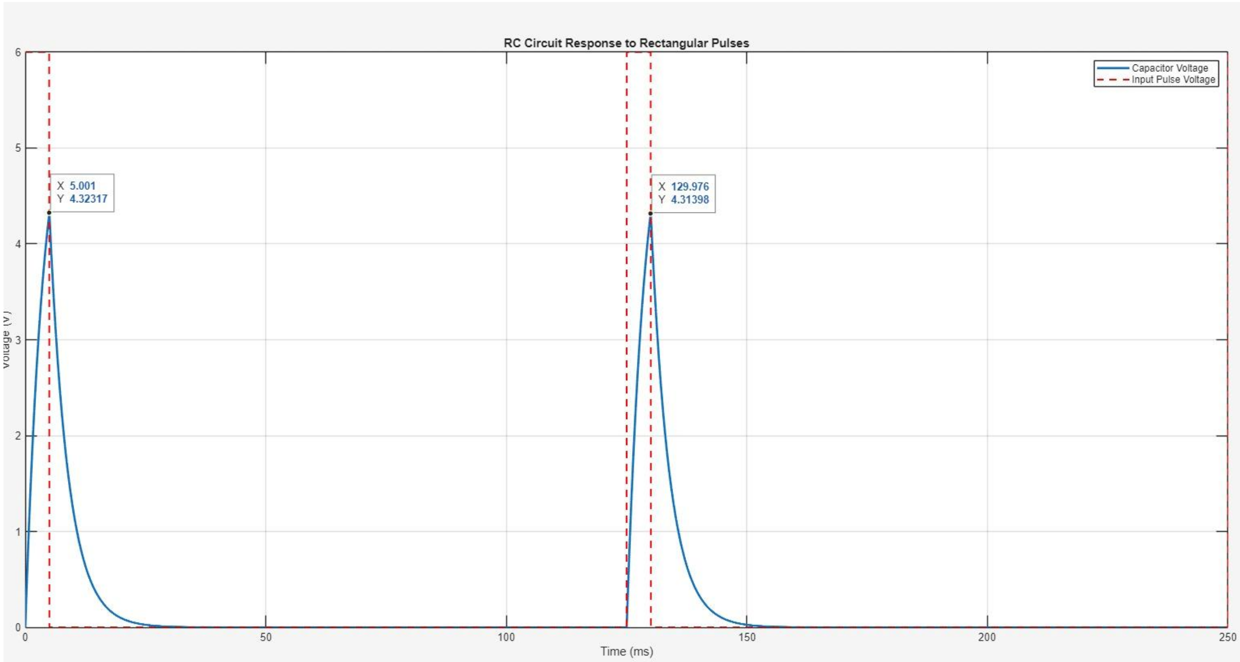

Figure B: MATLAB Plot of equations (5) and (6) for their respective time intervals. The code can be found in the appendix

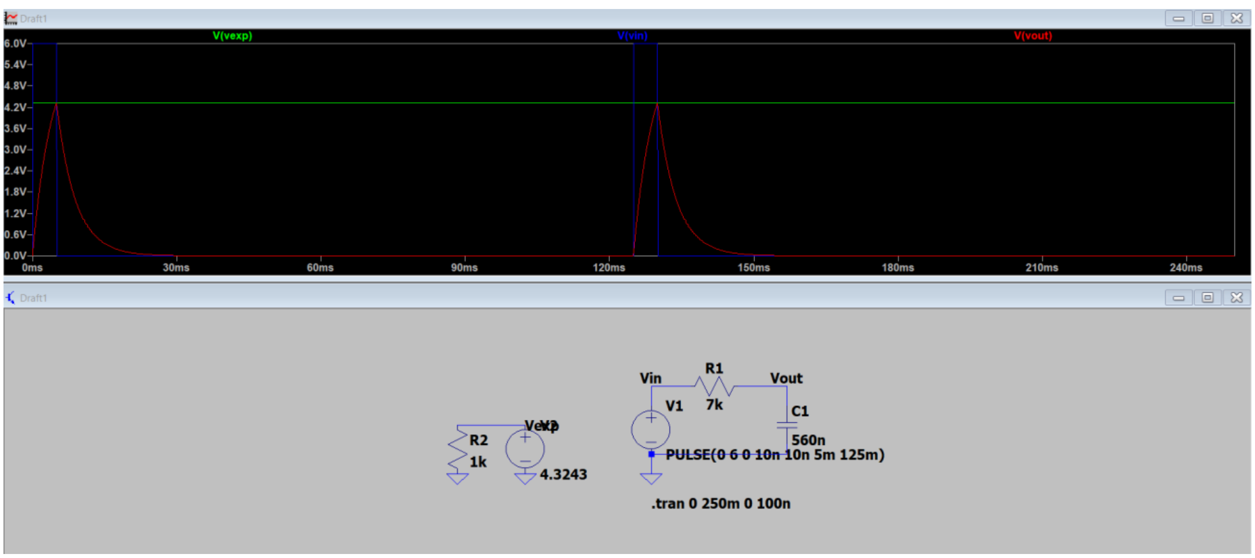

Figure C: LTSpice Circuit and plot of the rectangular pulse for the given parameters (Blue = Input voltage, Red = Output voltage, Green = Expected Voltage for pulse transition)

By inspection, it was found that both cases resulted in the same plot. During the pulse, the capacitor charges in the form of an inverse exponential and discharges after the pulse in the same manner. Since two different methods yielded the same result, we have gathered a strong basis to base the real-life experiment on for the rectangular pulse transient.

Preliminary Part 2 (Sinusoidal Pulse Response):

For the second part of this experiment, most parameters – such as the values of the resistor and capacitor, the initial voltage, pulse duration and period – were kept the same. Therefore, equation (4) could be used to gather the capacitor's transient output voltage. The source now consisted of a sinusoidal wave generator of magnitude three volts during the pulse duration with a period equal to the pulse duration for t<T_1 and an open circuit for T_1<t<T_0. Before substituting this pulse into equation (4), we can take advantage of the following general solution of the integral[2]:

$$\int e^{at}\,sin(bt)dt = \frac{e^{at}}{a^2+b^2}\left(\,a sin(\mathrm{bt}) - \mathrm{b} cos(\mathrm{bt})\,\right) + \mathrm{C} \quad (7)$$

This general solution and the initial condition of $V_{out}(0) = 0$ can be used in (4) to get:

$$V_{out}(t) = \; e^{-\frac{t}{RC}}\left(\frac{1}{RC}\int 3e^{\frac{t}{RC}}\sin\,(\omega t)\,dt + C\right)V = \frac{3}{1+(\omega RC)^2}\left(sin(\omega t) - \omega RCcos(\omega t) + \omega RCe^{\frac{-t}{RC}}\right)V \quad (8)$$

where $\omega = \frac{2\pi}{T_1}$. Substituting in the values for the components into (8), we finally get:

$$V_{out}(t) = 0.1186(\,sin(\omega t) - 4.9260cos(\omega t) + 4.9260e^{\frac{-t}{3.92ms}}\,) \quad (9)$$

which is the transient response of the capacitor to the sinusoidal pulse for the time interval 0<t<T_1 where t is in milliseconds.

For the interval of T_1<t<T_0, we can find the voltage at T_1 again, set it as the initial condition and re-evaluate the integral. At 5 ms, we get approximately -0.4211 Volts. Using this value and evaluating equation (6), we get:

$$V_{out}(t) = e^{-\frac{t}{3.92}}\left(\frac{1}{3.92}\int 0e^{\frac{t}{3.92}}\,dt + C\right)V = -0.4211e^{-\frac{t-T_1}{3.92ms}}V \quad (10)$$

for the time between the end of the current pulse and the start of the next pulse, which is the time interval of T_1<t<T_0 where t is in milliseconds. The same procedure that was used in part one of the preliminary work was utilized to plot the analytical and simulative results in MATLAB (Figure D, Appendix) and LTSpice (Figure E) respectively.

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[2] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products, 7th ed., New York, NY, USA: Academic Press,* 2007, ch. 2.3, eq. (2.3.3).

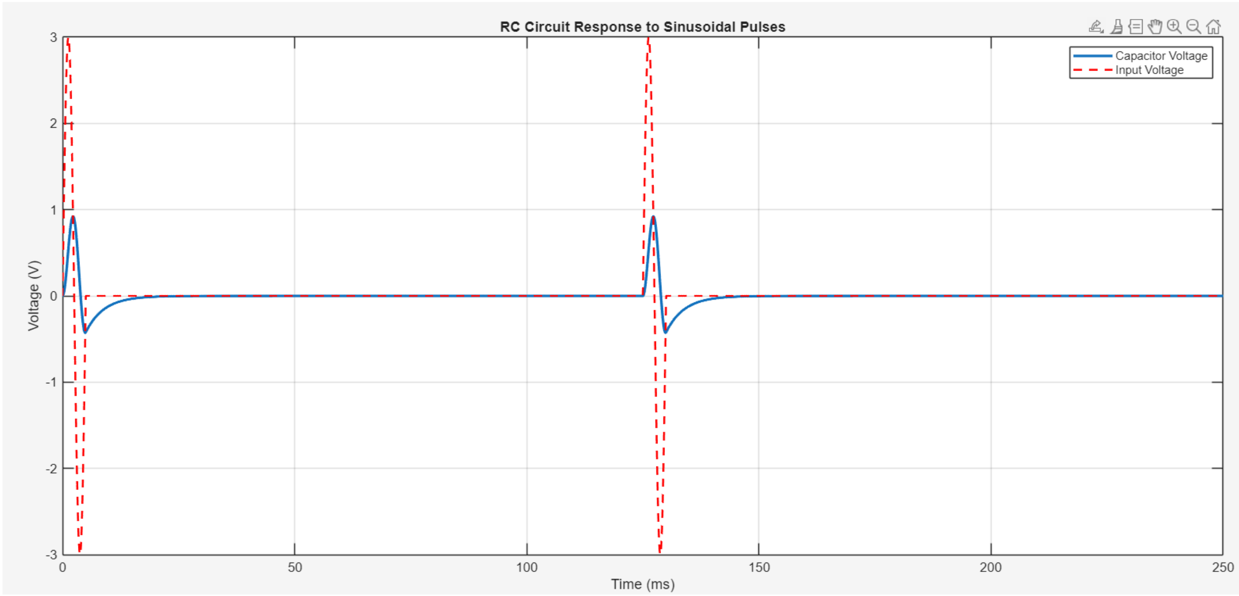

Figure D: MATLAB Plot of equations (9) and (10) for their respective time intervals. The code can be found in the appendix

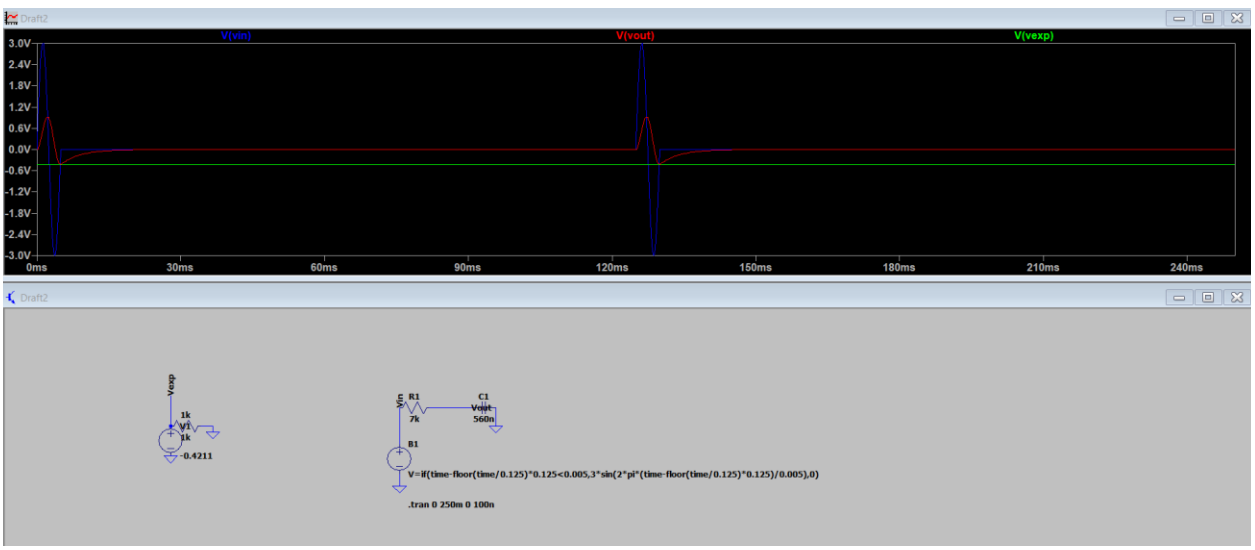

Figure E: LTSpice Circuit and plot of the sinusoidal pulse for the given parameters (Blue = Input voltage, Red = Output voltage, Green = Expected Voltage for pulse transition)

Both plots outputted the same result and matched our calculations. During the pulse, the capacitor voltage increases as the sum of a dampened exponential and a sinusoidal signal with a phase-shift in comparison to the source pulse voltage. Since the pulse is very short, the capacitor voltage cannot fully reach a steady state before discharging. Since both the simulative and the analytic case resulted in the same plot, we have a strong basis for our real-life experiment.

Hardware Implementation:

After the simulation and analytical results were gathered, the design was implemented on a breadboard (Figure 1.1). Two 100 Ohm resistors and one 6.8 kOhm resistor was connected in series to get a 7 kOhm resistance due to the unavailability of a direct 7 kOhm resistor. A 560 nanoFarrad capacitor was connected in series with the resistive column in accordance to the given design. The output of the signal generator was connected to the power rails of the breadboard. The positive rail was connected to the unused terminal of the resistive column and the ground rail was connected to the unused leg of the capacitor. For our two-channel measurements, two oscilloscope probes were connected to an oscilloscope. The positive clamp of the signal generator was probed and connected to channel two to view the input voltage, and the non-grounded leg of the capacitor was probed and connected to channel one to see the capacitor voltage. Both of the oscilloscope ground leads were connected to the grounded rail. Through these steps, the desired design was implemented.

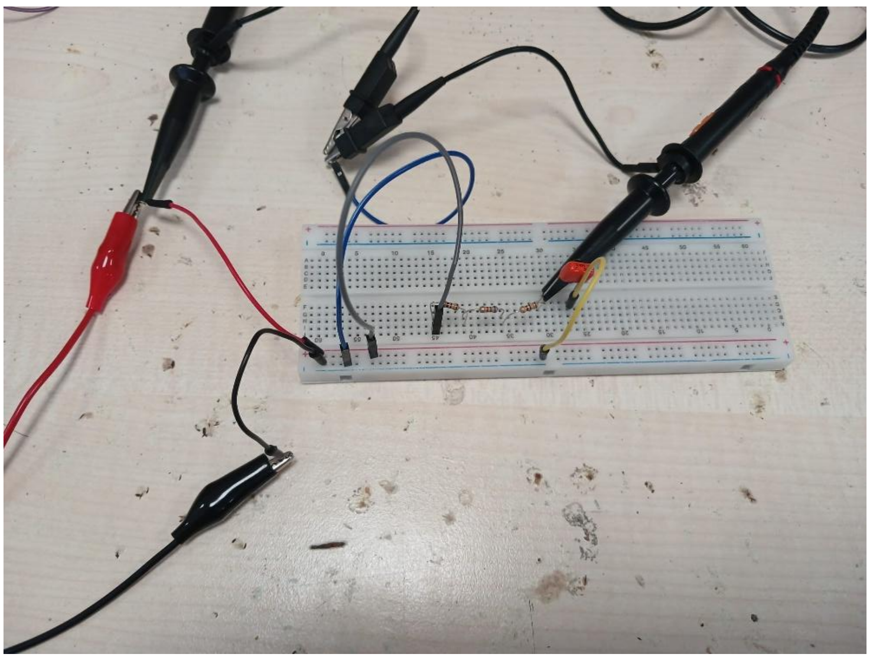

Figure 1.1: Implemented design with probes and power connected

After the physical design was completed, a coaxial cable was used to connect the TRIG OUTPUT of the signal generator to the EXT TRIG plug of the oscilloscope. For our measurements, the settings of the signal generator were set up in accordance to the provided Lab 1 document for the rectangular pulse section. When all respective steps were completed, a coaxial cable was connected to the FUNCTION output of the signal generator and the design was supplied with power. The waveforms were analyzed using the screen in a rough analysis, and cursors were placed on points of interest to measure voltage (Figure 1.2) at the end of the rectangular pulse and time (Figure 1.3) of the pulse duration and pulse period. Both points of data were recorded for later comparison.

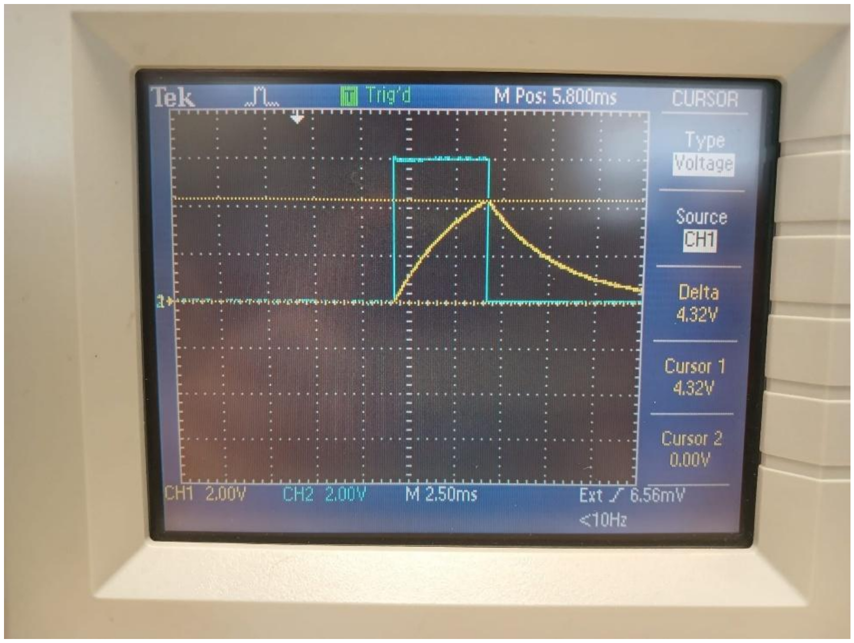

Figure 1.2: Input and Capacitor Voltage for Rectangular Pulses with transition capacitor voltage cursored

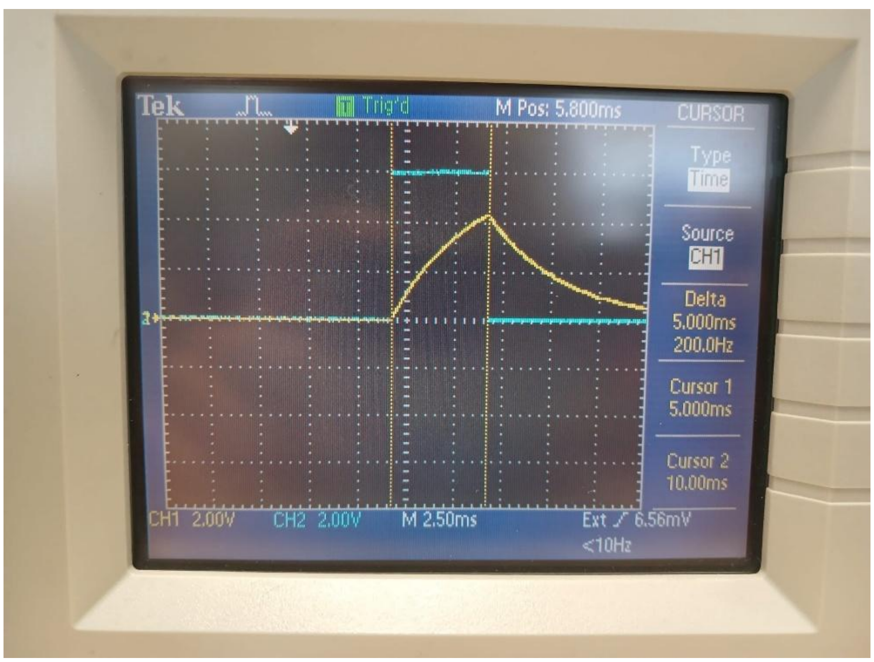

Figure 1.3: Input and Capacitor Voltage for Rectangular Pulses with pulse duration cursored

When the work concerning the rectangular pulse was completed, the signal generator's coaxial lead was disconnected. The signal generator's configuration was reset and the sinusoidal pulse excitation section of the Lab 1 document was followed to create a sinusoidal pulse. After the signal generator was set, the coaxial cable was reconnected and the waveforms of the oscilloscope were analyzed. After a rough waveform analysis by inspection, cursors were placed on points of interest to measure voltage (Figure 1.4) at the end for the sinusoidal pulse and time (Figure 1.5) of the pulse duration and pulse period. Both points of data were recorded for later comparison.

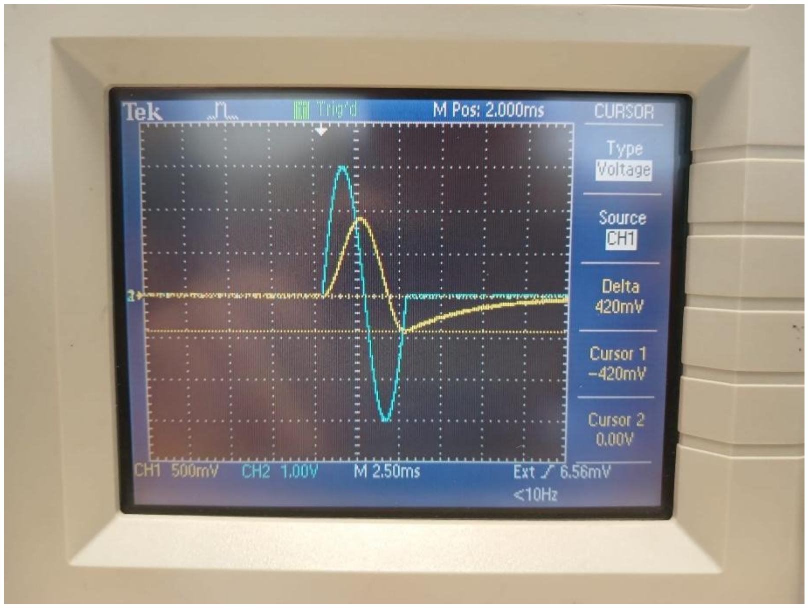

Figure 1.4: Input and Capacitor Voltage for Sinusoidal Pulses with transition capacitor voltage cursored

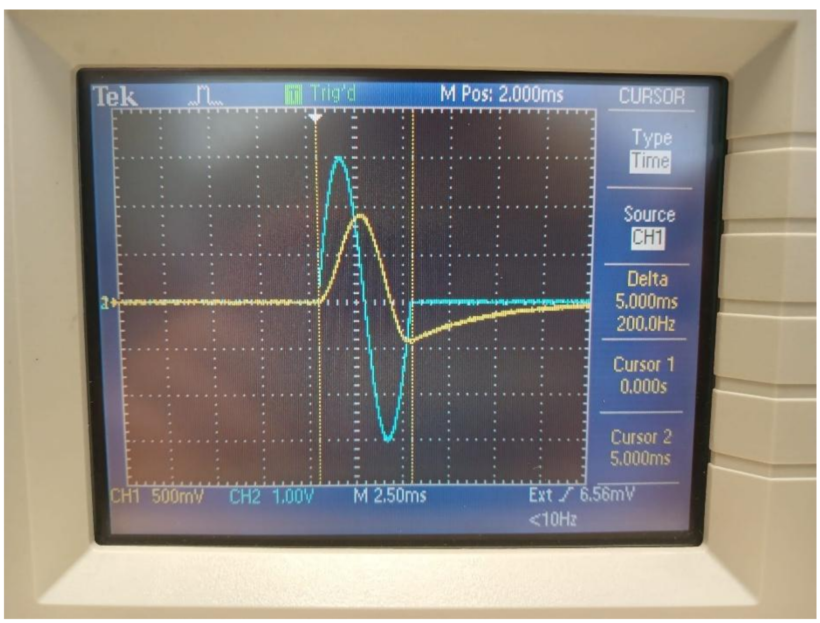

Figure 1.4: Input and Capacitor Voltage for Sinusoidal Pulses with pulse duration cursored

Using the data recorded in the hardware part of the lab and the values derived in the preliminary work, a table was formed to compare each data point and their errors (Table 1). Errors were calculated by comparing simulative and real-life results. Each data entry was looked at with their errors. An extremely small error value was measured for the hardware implementation when compared to the analytical section. These could be the result of human error during measurement and tolerances that come with each component. Due to the small error values, the experiment was deemed to be a success.

	Output Voltage (V)	Time of Measurement (ms)	Error (%)
MATLAB (Rectangular)	4.3232	5.001	-
LTSpice (Rectangular)	4.3170	5.007	-
Real-Life (Rectangular)	4.32	5	0.07
MATLAB (Sinusoidal)	-0.4210	5.001	-
LTSpice (Sinusoidal)	-0.4215	5.007	-
Real-Life (Sinusoidal)	-0.42	5	0.36

Table 1: Collection of analytical, simulative and real-life values and errors

Conclusion:

The purpose of this lab assignment was to understand the behavior of an RC circuit under the conditions of a pulse with a relatively large time between the pulses. First, the general solution of such systems was derived analytically and different parameters were adjusted to analyze specific cases, namely a rectangular pulse and a sinusoidal pulse. After these results of said systems were found, they were plotted using MATLAB. To compare the analytical solutions to a simulative case, the circuits were designed in LTSpice and a transient analysis was performed on them to find a very good approximation of the ideal cases. The respective voltages were compared and extremely similar plots were observed. Some errors were observed between the plots – less than one percent – and a possible reason for them could be the different methods between MATLAB and LTSpice when computing differing floating points and the timestep for both software being different, resulting in small differences. From these results, we have gathered very strong comparisons for the real-life experiment for the rectangular and sinusoidal pulses respectively.

After the simulative and analytical parts, the design was created in real life to get actual measurements. When the experiment was carried out, certain values were observed. These measurements were recorded and compared to the theoretical values to find the error. The error was calculated and was found to be extremely small relative to the measurements. Through this lab assignment, RC Voltage discharge, circuit analysis derivations and hardware implementation skills were gained and improved.

References:

I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 7th ed. Amsterdam, The Netherlands: Elsevier/Academic Press, 2007.

M. D. Greenberg, *Advanced Engineering Mathematics*, 2nd ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1988.

Appendix:

Code for Figure B (Rectangular pulses):

```
clear

clc

close all


% Parameters

R = 7e3;

C = 560e-9;

tau = R*C;


V = 6;

Ton = 5e-3;

T = 125e-3;


% Time axis: 0–250 ms (two pulses)

t = linspace(0, 250e-3, 5000);

tp = mod(t, T);   % time within each period


% Input pulse voltage

Vpulse = V * (tp < Ton);


% Capacitor voltage (analytical, piecewise)

Vc = zeros(size(t));


% Charging

idx_charge = tp < Ton;

Vc(idx_charge) = V * (1 - exp(-tp(idx_charge)/tau));


% Discharging

Vc_end = V * (1 - exp(-Ton/tau));
```

```
idx_discharge = tp >= Ton;
Vc(idx_discharge) = Vc_end * exp(-(tp(idx_discharge) - Ton)/tau);

% Plot
figure
plot(t*1e3, Vc, 'LineWidth', 2)         % Capacitor voltage
hold on
plot(t*1e3, Vpulse, 'r--', 'LineWidth', 1.5)  % Pulse voltage (red)
grid on

xlabel('Time (ms)')
ylabel('Voltage (V)')
title('RC Circuit Response to Rectangular Pulses')
legend('Capacitor Voltage', 'Input Pulse Voltage')
xlim([0 250])
```

Code for Figure D (Sinusoidal Pulses):

```
clear
clc
close all

% Parameters
R = 7e3;
C = 560e-9;
tau = R*C;

Ton = 5e-3;
T = 125e-3;
omega = 2*pi / Ton;

% Time axis
t = linspace(0, 250e-3, 5000);
```

```
tp = mod(t, T);   % local time in each period

% Input voltage
Vin = zeros(size(t));
idx_on = tp < Ton;
Vin(idx_on) = 3 * sin(omega * tp(idx_on));

% Capacitor voltage
Vc = zeros(size(t));

% During pulse
Vc(idx_on) = 0.1186 * ( sin(omega * tp(idx_on))  - 4.9260 * cos(omega * tp(idx_on))  + 4.9260 *
exp(-tp(idx_on) / tau) );

% Between pulses
Vc_end = 0.1186 * ( sin(omega * Ton) - 4.9260 * cos(omega * Ton) + 4.9260 * exp(-Ton / tau) );

idx_off = tp >= Ton;
Vc(idx_off) = Vc_end .* exp(-(tp(idx_off) - Ton) / tau);

% Plot
figure
plot(t*1e3, Vc, 'LineWidth', 2)
hold on
plot(t*1e3, Vin, 'r--', 'LineWidth', 1.5)
grid on

xlabel('Time (ms)')
ylabel('Voltage (V)')
title('RC Circuit Response to Sinusoidal Pulses')
legend('Capacitor Voltage', 'Input Voltage')
xlim([0 250])
```